Additional file 1: Supplementary results

Evidence before prediction: a configuration-driven framework for testing how molecular descriptors shape property models

Abbreviations. AP, average precision; AUROC, area under the receiver operating characteristic curve; BBBP, blood-brain barrier penetration; BACE-1, beta-site amyloid precursor protein cleaving enzyme 1; BCE, binary cross-entropy; CI, confidence interval; D-MPNN, directed message-passing neural network; ECFP4, extended-connectivity fingerprint with radius 2; ESOL, the aqueous-solubility benchmark; FFN, feed-forward network; GIN, graph isomorphism network; InChIKey, International Chemical Identifier key; L1, absolute-value regularisation; LARS, least angle regression; MAE, mean absolute error; MI, mutual information; MI–Pearson, mutual-information relevance with Pearson-correlation redundancy; MLP, multilayer perceptron; OOD, out of distribution, used here as a range-based descriptor diagnostic; RF, Random Forest; RMSE, root mean squared error; SD, standard deviation; SMILES, simplified molecular input line entry system; TDC, Therapeutics Data Commons.

Contents. Dataset overview, source-level target statistics, and provenance (Table S1a); fixed hyperparameters (Table S1b); scorecard diagnostics (Table S2); recommended versus random subsets and the full pool (Table S3); six-selector results and stability (Table S4, Fig. S1) and the selector comparisons that survive multiplicity correction (Table S4b); internal and external paired comparisons (Tables S5–S6); harmonised descriptor-routing results and their paired comparisons (Tables S7 and S7b); training-seed variance (Table S8); secondary metrics (Table S9); per-split primary scores (Table S10); Caco-2 advisor case study (Table S11); duplicate-controlled BBBP sensitivity analysis and its paired comparisons (Tables S12 and S12b); checkpoint protocol and result-file inventory (Table S13).

Conventions. Unless stated otherwise, scores are measured on the test partition of the ten saved split manifests, with split $s = 1, \ldots, 10$ corresponding to seed s. The duplicate controlled BBBP sensitivity analysis uses new manifests described in Table S12. Means are reported with population standard deviation (SD). Root mean squared error (RMSE) is better when lower; area under the receiver operating characteristic curve (AUROC) is better when higher. Paired differences are $\Delta_s = m_A(s) - m_B(s)$, and wins count the splits on which A is better. The descriptive 95% confidence interval uses $\bar{\Delta} \pm t_{0.975,9}\, s_\Delta/\sqrt{10}$, where s_Δ is the sample standard deviation of the paired differences; because training sets overlap, the corrected Nadeau–Bengio intervals in Tables S5–S6 and S7b inflate the variance by $1/k + n_{\text{test}}/n_{\text{train}}$ and are the intervals used for inference. Tables were generated from result files written by scripts in `benchmarks/` of the repository.

Dataset overview and provenance

Table S1a: Dataset overview, target definitions, source-level target distributions, and provenance. N is the number of records retained for this study after the stated preprocessing. Published target summaries are shown for the original dataset release and are not recomputed from the individual train, validation, or test partitions.

Dataset (N used)	Target / output	Source-level target distribution	Task / metric	Split	Original source
ESOL (1,128)	Measured aqueous solubility, $\log_{10}$ mol L^{-1}	Mean −3.05; SD 2.10; range $[-11.60, 1.58]$	Regression; RMSE	Scaffold	Delaney (2004)
FreeSolv (642)	Experimental hydration free energy, kcal mol^{-1}	Mean −3.80; SD 3.85; range $[-25.47, 3.43]$	Regression; RMSE	Random	Mobley et al. (2014); FreeSolv v0.52
Lipophilicity (4,200)	Octanol/water distribution coefficient, $\log D$ at pH 7.4	Mean 2.19; SD 1.20; assay range $[-1.5, 4.5]$	Regression; RMSE	Scaffold	Hersey et al. / ChEMBL (2015)
BBBP (2,039)	Blood–brain barrier permeability label	Original file: 1,567 positive (76.4%) and 483 negative of 2,050; 11 unparseable SMILES removed	Binary classification; AUROC	Scaffold	Martins et al. (2012)
BACE (1,513)	BACE-1 inhibitor/activity label	691 positive/active (45.7%) and 822 negative/inactive (54.3%)	Binary classification; AUROC	Scaffold	Subramanian et al. (2016)

The target summaries above are included to contextualise the prediction problems, not as evidence calculated from held-out labels. The counts used in the model benchmark follow the processed data described in the manuscript. In particular, the BBBP source release contains 2,050 records, whereas the primary benchmark contains 2,039 because 11 SMILES could not be parsed by RDKit. In the duplicate-controlled BBBP sensitivity analysis, the data composition and split manifests change again; those counts and class totals are reported separately in Table S12.

Hyperparameters

Table S1b: Fixed hyperparameters. No hyperparameter was tuned on test data; K^* in the routing experiment was the only quantity selected, on validation.

Model	Settings
ECFP4 + Random Forest	Morgan fingerprint, radius 2, 2,048 bits; 100 trees, scikit-learn defaults otherwise
RDKit 2D + Random Forest	210 RDKit 2D descriptors (RDKit 2023.09.6), non-finite values set to 0; 100 trees
Descriptor MLP (reference)	$210 \to 128 \to 64 \to 1$, LayerNorm, dropout 0.2; AdamW (lr 10^{-3}, weight decay 10^{-4}), batch 64, 50 epochs, Huber or BCE-with-logits loss, checkpoint on validation loss
Framework role ablation (`direct_only`, `residual_only`, `direct_hybrid`, `prior_residual`)	D-MPNN, hidden 128, depth 3, sum aggregation; descriptor encoder $210 \to 64$ with LayerNorm; head $192 \to 64 \to 1$; dropout 0.2; AdamW (lr 10^{-3}, weight decay 10^{-4}), batch 64, 40 epochs, checkpoint on validation loss; standard scaling fitted on training data
Prior (`prior_only`, `prior_residual`)	Six descriptors (MolWt, MolLogP, TPSA, NumHDonors, NumHAcceptors, NumRotatableBonds); ridge regression ($\alpha = 1$) or logistic regression ($C = 1$, log-odds from `decision_function`), fitted on training data and frozen
Harmonised routing experiment	Same architecture and optimiser as the role ablation; $K \in \{10, 20, 50\}$; all route checkpoints selected by minimum validation loss; K^* selected using the primary validation metric at that checkpoint; random routes averaged over ten draws per split
Checkpoint audit	Completed across five datasets; 250 model runs logged in `routing_checkpoint_audit.csv`; dataset-level outputs are listed in Table S13
Chemprop v2.3.1	Bond message passing, hidden size 300, depth 3, dropout 0, mean aggregation, batch normalisation; FFN with one hidden layer of 300; default learning-rate schedule; batch 64; up to 50 epochs, early stopping on validation loss (patience 10); target and descriptor scaling fitted on training data
Selector comparison	MI: $k = 3$ neighbours; MI–Pearson: MI relevance minus mean |Pearson| redundancy; RF importance: 100 trees, max depth 12; L1: LARS–lasso entry order (regression) or |coefficient| of L1-logistic regression, $C = 1$ (classification); L1 feature scaling fitted on training data only; downstream Random Forest with 100 trees

Scorecard diagnostics

Table S2: Range-based covariate shift. (a) Illustrative scaffold split 1 versus a random 80/10/10 split of the same data. OOD descriptors: descriptors with any test value outside the training range. Evidence classes in this case study follow a simplified rule of the audit script: out-of-range or zero training support → OOD/unsupported; $|r| > 0.85$ with an earlier feature → redundant; training $|\rho_{\text{Spearman}}| \geq 0.15$ → supported; otherwise weak. (b) The ten BBBP scaffold splits with the number of out-of-range descriptors, the number of test molecules affected, and the test AUROC of four role-ablation models.

(a)

Dataset	Split	OOD descriptors	Test molecules affected	OOD/unsupported	Redundant	Supported	Weak
ESOL	Scaffold	20 (9.5%)	7.1%	39	48	34	89
	Random	8 (3.8%)	5.3%	27	51	35	97
Lipophilicity	Scaffold	11 (5.2%)	3.1%	21	43	25	121
	Random	7 (3.3%)	1.7%	17	45	23	125

(b)

Split	OOD descriptors	Test molecules affected	`direct_only`	`direct_hybrid`	`residual_only`	`prior_residual`
1	34 (16.2%)	10 (4.9%)	0.9320	0.9367	0.9162	0.9375
2	5 (2.4%)	4 (2.0%)	0.8721	0.9118	0.8265	0.8277
3	18 (8.6%)	7 (3.4%)	0.9429	0.9507	0.9270	0.9291
4	8 (3.8%)	9 (4.4%)	0.9214	0.9110	0.9243	0.9108
5	12 (5.7%)	5 (2.5%)	0.8819	0.8958	0.9386	0.9331
6	16 (7.6%)	11 (5.4%)	0.8174	0.8833	0.8850	0.9109
7	4 (1.9%)	5 (2.5%)	0.9496	0.9659	0.9496	0.9455
8	19 (9.0%)	18 (8.8%)	0.8578	0.8671	0.9413	0.9302
9	6 (2.9%)	4 (2.0%)	0.9319	0.9280	0.9258	0.8854
10	28 (13.3%)	3 (1.5%)	0.9240	0.9115	0.9097	0.8872

Across all ten scaffold splits, the mean number of out-of-range descriptors was 19.3 ± 12.7 for ESOL ($9.2 \pm 6.1\%$ of the 210 descriptors), 20.8 ± 14.8 for Lipophilicity ($9.9 \pm 7.1\%$), and 15.0 ± 9.6 for BBBP ($7.1 \pm 4.6\%$). The mean fraction of test molecules affected was $6.3 \pm 2.5\%$, $3.7 \pm 0.7\%$, and $3.7 \pm 2.1\%$, respectively. The scaffold-versus-random values in Table S2(a) are a split-1 illustration, not averages across split manifests.

Recommended subsets versus random subsets and the full pool

Table S3: Framework-recommended top-K subsets versus random K-subsets (ten draws per split, averaged) and versus the full 210-descriptor pool, with a Random Forest downstream. All entries are paired on the same 10 splits. $\bar{\Delta}$ is the mean paired difference, recommended minus the comparator, so a negative value in panel (a) and a positive value in panel (b) mean that the recommended subset scores worse than the comparator (RMSE for ESOL and FreeSolv; AUROC for BBBP). p_{H} is Holm-adjusted within each dataset across the eight comparisons of the two panels ($m = 8$); $^{*}p_{\mathrm{H}} < 0.05$, $^{**}p_{\mathrm{H}} < 0.01$. Against random subsets of the same size, the recommended subsets score significantly better at ESOL $K = 5$, ESOL $K = 10$, ESOL $K = 20$, ESOL $K = 50$, FreeSolv $K = 5$, FreeSolv $K = 10$, FreeSolv $K = 20$, and BBBP $K = 5$ and nowhere worse; the remaining 4 of the 12 dataset-budget cells do not separate. Against the full 210-descriptor pool, the recommended subsets are significantly worse at ESOL $K = 5$ and ESOL $K = 10$, and no other cell separates (10 of 12 cells inconclusive). Random subsets average ten draws per split, so differences against them carry the draw noise of the comparator. Contrast this table with the corresponding rows of Table S7, where the same recommendation is routed into a graph hybrid rather than used directly; the two experiments ask different questions and are not expected to agree.

(a) Recommended versus random subsets

Dataset	K	$\bar{\Delta}$	Corrected 95% interval	p_{H}	Wins of recommended
ESOL	5	-0.745	$[-0.894, -0.596]$	$< 0.001^{**}$	10/10
	10	-0.419	$[-0.526, -0.313]$	$< 0.001^{**}$	10/10
	20	-0.204	$[-0.307, -0.102]$	0.006^{**}	10/10
	50	-0.116	$[-0.165, -0.068]$	0.002^{**}	10/10
FreeSolv	5	-1.546	$[-1.924, -1.169]$	$< 0.001^{**}$	10/10
	10	-1.009	$[-1.223, -0.794]$	$< 0.001^{**}$	10/10
	20	-0.567	$[-0.785, -0.350]$	0.001^{**}	10/10
	50	-0.300	$[-0.515, -0.084]$	0.059	10/10
BBBP	5	$+0.096$	$[0.039, 0.152]$	0.032^{*}	10/10
	10	$+0.043$	$[0.007, 0.078]$	0.140	8/10
	20	$+0.015$	$[-0.006, 0.036]$	0.551	8/10
	50	$+0.004$	$[-0.005, 0.014]$	0.671	4/10

(b) Recommended versus the full 210-descriptor pool

Dataset	K	$\bar{\Delta}$	Corrected 95% interval	p_{H}	Wins of recommended
ESOL	5	$+0.063$	$[0.031, 0.094]$	0.006^{**}	0/10
	10	$+0.080$	$[0.048, 0.111]$	0.002^{**}	0/10
	20	$+0.055$	$[0.002, 0.107]$	0.088	0/10
	50	$+0.010$	$[-0.016, 0.036]$	0.389	5/10
FreeSolv	5	$+0.321$	$[0.062, 0.580]$	0.082	1/10
	10	$+0.151$	$[-0.022, 0.324]$	0.238	2/10
	20	$+0.092$	$[-0.042, 0.227]$	0.310	2/10
	50	-0.019	$[-0.097, 0.058]$	0.584	5/10
BBBP	5	-0.048	$[-0.084, -0.012]$	0.097	1/10
	10	-0.023	$[-0.050, 0.003]$	0.376	2/10
	20	-0.009	$[-0.023, 0.006]$	0.629	2/10
	50	-0.001	$[-0.010, 0.007]$	0.766	3/10

Six feature selectors

Table S4: Six feature selectors with a Random Forest downstream, on the ten saved splits. (a) Mean test score per selector and budget (RMSE for ESOL, FreeSolv and Lipophilicity, AUROC for BBBP and BACE); bold marks the best mean within each dataset and budget. (b) Mean pairwise Jaccard index of the selected sets across splits, averaged over $K \in \{5, 10, 20, 50\}$. The selector previously labelled mRMR is named MI–Pearson here because its redundancy term is mean absolute Pearson correlation rather than mutual information. The table reports selectors as the learner's only input; Table S7 instead routes the recommended subset into a graph hybrid.

(a) Test score by selector and budget

Dataset (metric)	Selector	$K = 5$	$K = 10$	$K = 20$	$K = 50$
ESOL (RMSE ↓)	Framework recommender	0.700	0.716	0.692	0.647
	L1 (lasso / L1-logistic)	0.734	0.729	0.704	0.675
	MI–Pearson	0.711	0.677	0.658	0.659
	Mutual information	0.682	0.661	0.663	0.651
	RF impurity importance	**0.678**	**0.633**	**0.640**	**0.640**
	Spearman top-K	0.718	0.668	0.664	0.648
FreeSolv (RMSE ↓)	Framework recommender	1.410	1.240	1.182	**1.070**
	L1 (lasso / L1-logistic)	1.724	1.350	1.245	1.229
	MI–Pearson	1.281	1.263	1.183	1.106
	Mutual information	**1.243**	1.192	1.168	1.124
	RF impurity importance	1.277	**1.108**	**1.095**	1.077
	Spearman top-K	1.456	1.222	1.170	1.074
Lipophilicity (RMSE ↓)	Framework recommender	0.941	0.850	0.788	0.768
	L1 (lasso / L1-logistic)	0.964	0.905	0.821	0.784
	MI–Pearson	0.880	0.831	0.800	0.779
	Mutual information	**0.857**	**0.814**	0.783	0.763
	RF impurity importance	0.922	0.822	**0.780**	**0.759**
	Spearman top-K	0.984	0.857	0.798	0.782
BBBP (AUROC ↑)	Framework recommender	0.864	0.889	0.904	0.911
	L1 (lasso / L1-logistic)	0.832	0.891	0.906	**0.919**
	MI–Pearson	**0.884**	**0.906**	**0.912**	0.914
	Mutual information	0.879	0.899	0.904	0.916
	RF impurity importance	0.860	0.886	0.903	0.909
	Spearman top-K	0.853	0.875	0.899	0.907
BACE (AUROC ↑)	Framework recommender	0.758	0.780	0.796	0.818
	L1 (lasso / L1-logistic)	0.716	0.764	0.797	**0.820**
	MI–Pearson	**0.786**	0.793	0.808	0.807
	Mutual information	0.783	0.793	0.807	0.809
	RF impurity importance	0.773	**0.796**	**0.810**	0.818
	Spearman top-K	0.748	0.763	0.772	0.798

(b) Mean pairwise Jaccard stability across splits

Selector	ESOL	FreeSolv	Lipophilicity	BBBP	BACE
Framework recommender	0.78	0.84	0.79	0.76	0.62
L1 (lasso / L1-logistic)	0.50	0.38	0.59	0.51	0.39
MI–Pearson	0.56	0.78	0.43	0.46	0.61
Mutual information	0.86	0.88	0.63	0.82	0.83
RF impurity importance	0.67	0.71	0.70	0.62	0.57
Spearman top-K	0.96	0.92	0.82	0.85	0.86

Table S4b: Selector comparisons that survive Holm correction within a dataset (family $m = 20$: the five other selectors against the framework selector at four budgets, on the same ten splits as Table S4). Of the 100 selector comparisons, 2 survive: `rf_importance` $K = 10$ vs framework $K = 10$ on ESOL and mutual_information $K = 5$ vs framework $K = 5$ on Lipophilicity. On the other endpoints no selector separates from the framework selector once multiplicity is controlled, which is consistent with the mean-rank table rather than in tension with it.

Dataset	Comparison (A vs B)	$\bar{\Delta} \pm$ SD	Corrected 95% interval	p	p_{H}	Wins of A
ESOL	`rf_importance` $K = 10$ vs framework $K = 10$	-0.084 ± 0.034	$[-0.122, -0.046]$	< 0.001	0.015^{*}	10/10
FreeSolv	no comparison survives the correction					
Lipophilicity	mutual_information $K = 5$ vs framework $K = 5$	-0.084 ± 0.025	$[-0.113, -0.055]$	< 0.001	0.002^{**}	10/10
BBBP	no comparison survives the correction					
BACE	no comparison survives the correction					

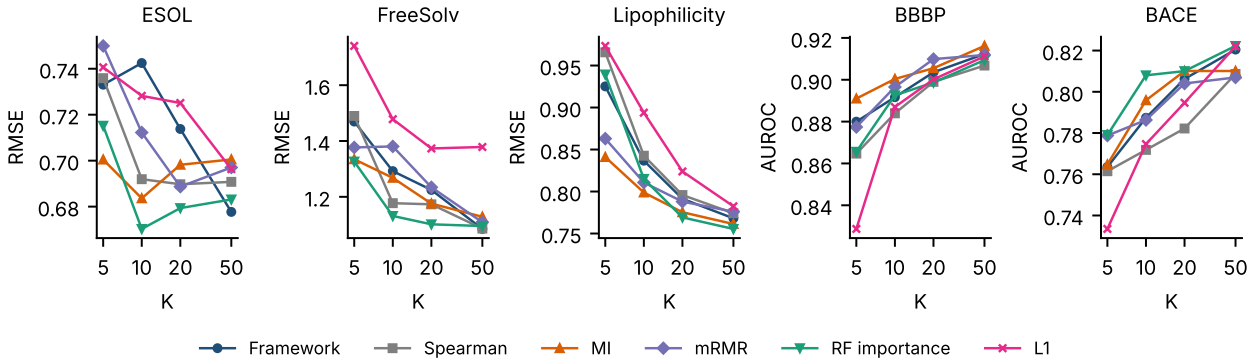


Figure S1: Mean test score of the six selectors as a function of the descriptor budget K (Random Forest downstream). Lower is better for RMSE, higher for AUROC. The former mRMR implementation is labelled MI–Pearson.

Paired comparisons

Table S5: Internal paired comparisons on the main models. Hybrid: `direct_hybrid`; graph only: `residual_only`; prior/residual: `prior_residual`. All entries are paired on the same 10 saved splits, and win counts are out of 10. $\bar{\Delta}$ is the mean paired difference $m_A - m_B$ with the population SD of the differences; wins count the splits on which A was better. Both intervals are two-sided t intervals with 9 degrees of freedom: the nominal interval uses the paired standard error, and the corrected interval inflates the variance by $1/k + n_{\text{test}}/n_{\text{train}}$ to reflect the overlap of the training partitions (Nadeau–Bengio). p is the corrected two-sided p-value and p_{H} is Holm-adjusted within this table ($m = 20$); $^{*}p_{\text{H}} < 0.05$, $^{**}p_{\text{H}} < 0.01$. The comparisons that survive Holm correction are Hybrid vs RDKit 2D + RF on Lipophilicity and Hybrid vs RDKit 2D + RF on BACE; every other entry is reported as inconclusive.

Dataset (metric)	Comparison (A vs B)	$\bar{\Delta} \pm$ SD	Nominal 95% interval	Corrected 95% interval	p	p_{H}	Wins of A
ESOL (RMSE)	Hybrid vs RDKit 2D + RF	-0.014 ± 0.0622	$[-0.061, 0.033]$	$[-0.085, 0.056]$	0.657	1.000	5/10
	Hybrid vs Graph only	-0.001 ± 0.0844	$[-0.064, 0.063]$	$[-0.096, 0.095]$	0.986	1.000	7/10
	Prior/residual vs Graph only	$+0.057 \pm 0.0990$	$[-0.018, 0.132]$	$[-0.055, 0.169]$	0.278	0.899	4/10
	Prior/residual vs Hybrid	$+0.058 \pm 0.0888$	$[-0.009, 0.125]$	$[-0.043, 0.158]$	0.225	0.899	2/10
FreeSolv (RMSE)	Hybrid vs RDKit 2D + RF	$+0.010 \pm 0.1737$	$[-0.121, 0.141]$	$[-0.187, 0.207]$	0.913	1.000	5/10
	Hybrid vs Graph only	-0.152 ± 0.2523	$[-0.342, 0.039]$	$[-0.438, 0.135]$	0.262	1.000	6/10
	Prior/residual vs Graph only	-0.076 ± 0.2818	$[-0.288, 0.137]$	$[-0.396, 0.244]$	0.605	1.000	7/10
	Prior/residual vs Hybrid	$+0.076 \pm 0.3086$	$[-0.157, 0.309]$	$[-0.275, 0.426]$	0.636	1.000	4/10
Lipophilicity (RMSE)	Hybrid vs RDKit 2D + RF	-0.142 ± 0.0323	$[-0.167, -0.118]$	$[-0.179, -0.106]$	< 0.001	$< 0.001^{**}$	10/10
	Hybrid vs Graph only	-0.001 ± 0.0228	$[-0.018, 0.016]$	$[-0.027, 0.025]$	0.943	1.000	5/10
	Prior/residual vs Graph only	$+0.002 \pm 0.0332$	$[-0.023, 0.028]$	$[-0.035, 0.040]$	0.885	1.000	6/10
	Prior/residual vs Hybrid	$+0.003 \pm 0.0285$	$[-0.018, 0.025]$	$[-0.029, 0.036]$	0.821	1.000	6/10
BBBP (AUROC)	Hybrid vs RDKit 2D + RF	$+0.004 \pm 0.0283$	$[-0.017, 0.025]$	$[-0.028, 0.036]$	0.783	1.000	5/10
	Hybrid vs Graph only	$+0.002 \pm 0.0400$	$[-0.028, 0.032]$	$[-0.043, 0.047]$	0.931	1.000	6/10
	Prior/residual vs Graph only	-0.005 ± 0.0184	$[-0.019, 0.009]$	$[-0.026, 0.016]$	0.625	1.000	4/10
	Prior/residual vs Hybrid	-0.006 ± 0.0401	$[-0.037, 0.024]$	$[-0.052, 0.039]$	0.755	1.000	4/10
BACE (AUROC)	Hybrid vs RDKit 2D + RF	-0.037 ± 0.0182	$[-0.051, -0.023]$	$[-0.058, -0.017]$	0.003	0.011^{*}	0/10
	Hybrid vs Graph only	-0.017 ± 0.0357	$[-0.044, 0.010]$	$[-0.057, 0.024]$	0.368	1.000	2/10
	Prior/residual vs Graph only	-0.019 ± 0.0495	$[-0.057, 0.018]$	$[-0.075, 0.037]$	0.457	1.000	4/10
	Prior/residual vs Hybrid	-0.002 ± 0.0478	$[-0.038, 0.034]$	$[-0.057, 0.052]$	0.924	1.000	6/10

Table S6: Framework minus Chemprop v2.3.1 on matched splits, for the graph only model and for the graph model with the 210 descriptors. Conventions follow Table S5; all entries are paired on the same 10 splits and p_{H} is Holm-adjusted within this table ($m = 10$). No comparison in this table survives Holm correction, so every entry is reported as inconclusive. Adding the 210 descriptors raises the mean score in both implementations on FreeSolv, Lipophilicity, and BBBP, lowers it in both on BACE, and disagrees between implementations on ESOL. Because a comparison between the two implementations at this sample size is underpowered, the agreement is descriptive.

Dataset (metric)	Model pair	$\bar{\Delta} \pm$ SD	Nominal 95% interval	Corrected 95% interval	p	p_{H}	Wins of framework
ESOL (RMSE)	graph only	-0.167 ± 0.1261	$[-0.262, -0.072]$	$[-0.310, -0.024]$	0.026	0.053	9/10
ESOL (RMSE)	graph + 210 descriptors	-0.221 ± 0.6143	$[-0.685, 0.242]$	$[-0.917, 0.474]$	0.490	0.490	7/10
FreeSolv (RMSE)	graph only	$+0.073 \pm 0.2786$	$[-0.137, 0.283]$	$[-0.243, 0.389]$	0.615	0.615	3/10
FreeSolv (RMSE)	graph + 210 descriptors	$+0.153 \pm 0.1913$	$[0.009, 0.298]$	$[-0.064, 0.370]$	0.145	0.290	1/10
Lipophilicity (RMSE)	graph only	-0.034 ± 0.0775	$[-0.092, 0.025]$	$[-0.121, 0.054]$	0.408	0.581	7/10
Lipophilicity (RMSE)	graph + 210 descriptors	-0.033 ± 0.0593	$[-0.078, 0.011]$	$[-0.100, 0.034]$	0.290	0.581	6/10
BBBP (AUROC)	graph only	$+0.003 \pm 0.0378$	$[-0.025, 0.032]$	$[-0.039, 0.046]$	0.859	1.000	4/10
BBBP (AUROC)	graph + 210 descriptors	-0.007 ± 0.0209	$[-0.023, 0.009]$	$[-0.031, 0.017]$	0.529	1.000	5/10
BACE (AUROC)	graph only	-0.006 ± 0.0294	$[-0.028, 0.017]$	$[-0.039, 0.028]$	0.710	1.000	3/10
BACE (AUROC)	graph + 210 descriptors	-0.007 ± 0.0208	$[-0.023, 0.009]$	$[-0.031, 0.016]$	0.512	1.000	4/10

Table S7: Harmonised evidence-guided descriptor routing. Entries are test RMSE for ESOL, FreeSolv, and Lipophilicity, and test AUROC for BBBP and BACE; values are mean ± population SD over ten saved splits. All route checkpoints are selected by minimum validation loss, and the recommended route's K^* is selected using the primary validation metric at that checkpoint. Spearman and MI use the same split-specific K^*; the random route averages ten random-subset draws per split. Bold text marks the best mean in each row. The full checkpoint-level audit covers 250 model runs; see Additional file 1, Table S13.

Dataset	Metric	Graph only	All 210	Recommender	Spearman	MI	Random
ESOL	RMSE ↓	0.638 ± 0.074	$\mathbf{0.614 \pm 0.039}$	0.620 ± 0.078	0.620 ± 0.075	0.622 ± 0.062	0.631 ± 0.057
FreeSolv	RMSE ↓	1.205 ± 0.164	$\mathbf{1.038 \pm 0.214}$	1.147 ± 0.140	1.134 ± 0.156	1.168 ± 0.180	1.149 ± 0.141
Lipophilicity	RMSE ↓	0.607 ± 0.031	0.606 ± 0.019	$\mathbf{0.581 \pm 0.041}$	0.592 ± 0.033	0.599 ± 0.032	0.591 ± 0.027
BBBP	AUROC ↑	0.914 ± 0.034	$\mathbf{0.916 \pm 0.029}$	0.915 ± 0.023	0.913 ± 0.020	0.910 ± 0.018	0.910 ± 0.020
BACE	AUROC ↑	$\mathbf{0.798 \pm 0.038}$	0.786 ± 0.033	0.797 ± 0.036	0.786 ± 0.059	0.784 ± 0.051	0.792 ± 0.042

Harmonised evidence-guided descriptor routing

All compact-route checkpoints were selected using minimum validation loss. For each split, the recommended route's budget $K^* \in \{10, 20, 50\}$ was selected by the primary validation metric evaluated at the selected checkpoint. Spearman and mutual-information controls reused that split-specific budget, while the random route was averaged over ten draws per split. The graph-only and all-210-descriptor references used validation-loss-selected checkpoints. All five endpoints have now been rerun under this common rule. Table S7 summarizes the test results; the full checkpoint-level log contains 250 model runs and is archived at `benchmarks/results/routing_checkpoint_audit.csv`. Individual dataset outputs are listed in Table S13.

Table S7b: Paired comparisons for the harmonised descriptor routing: the recommender route minus each reference route, on the 10 splits of Table S7. Negative values on the regression tasks and positive values on the classification tasks mean that the recommender route is better. p_{H} is Holm-adjusted within each dataset ($m = 5$); $^{*}p_{\mathrm{H}} < 0.05$, $^{**}p_{\mathrm{H}} < 0.01$. No comparison in this table survives Holm correction, so every entry is reported as inconclusive. The smallest corrected p-value in this table belongs to Recommender route vs Graph only on Lipophilicity ($p = 0.121$, $p_{\mathrm{H}} = 0.605$).

Dataset	Recommender minus	$\bar{\Delta} \pm$ SD	Corrected 95% interval	p	p_{H}	Wins
ESOL	Graph only	-0.017 ± 0.053	$[-0.077, 0.042]$	0.528	1.000	7/10
ESOL	All 210 descriptors	$+0.006 \pm 0.074$	$[-0.077, 0.089]$	0.875	1.000	5/10
ESOL	Spearman control	-0.000 ± 0.035	$[-0.040, 0.040]$	0.997	1.000	4/10
ESOL	Mutual-information control	-0.001 ± 0.061	$[-0.070, 0.068]$	0.964	1.000	5/10
ESOL	Random subsets	-0.011 ± 0.038	$[-0.054, 0.033]$	0.597	1.000	6/10
FreeSolv	Graph only	-0.058 ± 0.108	$[-0.180, 0.065]$	0.313	1.000	7/10
FreeSolv	All 210 descriptors	$+0.108 \pm 0.135$	$[-0.045, 0.262]$	0.145	0.724	3/10
FreeSolv	Spearman control	$+0.013 \pm 0.101$	$[-0.101, 0.127]$	0.804	1.000	4/10
FreeSolv	Mutual-information control	-0.021 ± 0.098	$[-0.132, 0.090]$	0.679	1.000	6/10
FreeSolv	Random subsets	-0.003 ± 0.053	$[-0.063, 0.058]$	0.919	1.000	4/10
Lipophilicity	Graph only	-0.026 ± 0.030	$[-0.061, 0.008]$	0.121	0.605	8/10
Lipophilicity	All 210 descriptors	-0.025 ± 0.032	$[-0.062, 0.011]$	0.154	0.605	9/10
Lipophilicity	Spearman control	-0.011 ± 0.018	$[-0.031, 0.009]$	0.244	0.605	6/10
Lipophilicity	Mutual-information control	-0.018 ± 0.023	$[-0.043, 0.008]$	0.150	0.605	8/10
Lipophilicity	Random subsets	-0.010 ± 0.023	$[-0.035, 0.016]$	0.400	0.605	9/10
BBBP	Graph only	$+0.001 \pm 0.038$	$[-0.043, 0.044]$	0.965	1.000	6/10
BBBP	All 210 descriptors	-0.001 ± 0.014	$[-0.017, 0.015]$	0.881	1.000	6/10
BBBP	Spearman control	$+0.002 \pm 0.026$	$[-0.027, 0.032]$	0.863	1.000	5/10
BBBP	Mutual-information control	$+0.005 \pm 0.019$	$[-0.016, 0.027]$	0.578	1.000	6/10
BBBP	Random subsets	$+0.005 \pm 0.015$	$[-0.011, 0.022]$	0.490	1.000	8/10
BACE	Graph only	-0.001 ± 0.029	$[-0.034, 0.031]$	0.924	1.000	4/10
BACE	All 210 descriptors	$+0.011 \pm 0.019$	$[-0.011, 0.033]$	0.288	1.000	7/10
BACE	Spearman control	$+0.011 \pm 0.038$	$[-0.031, 0.054]$	0.571	1.000	7/10
BACE	Mutual-information control	$+0.013 \pm 0.031$	$[-0.023, 0.048]$	0.441	1.000	5/10
BACE	Random subsets	$+0.005 \pm 0.017$	$[-0.015, 0.024]$	0.610	1.000	6/10

Training-seed variance

Table S8: Test AUROC on BBBP split 1 with three training seeds. SD: population SD over seeds. Between-split SDs over the ten benchmark splits for comparison: graph only 0.034, all 210 descriptors 0.029, prior/residual 0.034.

Model	Seed 42	Seed 123	Seed 999	Mean	SD
Graph only	0.9334	0.9405	0.9259	0.9333	0.0059
Graph + 210 descriptors	0.9449	0.9380	0.9383	0.9404	0.0032
Graph + recommended ($K = 20$)	0.9387	0.9267	0.9428	0.9361	0.0069
Prior + graph residual	0.9472	0.9353	0.9150	0.9325	0.0133

Secondary metrics and per-split scores

Table S9: Primary and secondary metrics for the four reference models (mean ± SD over ten saved splits). For the regression endpoints the primary metric is RMSE and the secondary metrics are MAE and R^2; for the classification endpoints the primary metric is AUROC and the secondary metric is AUPRC (area under the precision-recall curve).

Dataset	Model	Primary metric	MAE / AUPRC	R^2
ESOL	Descriptor MLP	0.623 ± 0.040	0.436 ± 0.027	0.886 ± 0.028
	Dummy	1.894 ± 0.197	1.491 ± 0.121	-0.017 ± 0.018
	ECFP4 + RF	1.352 ± 0.093	1.037 ± 0.056	0.470 ± 0.098
	RDKit 2D + RF	0.637 ± 0.061	0.454 ± 0.037	0.882 ± 0.027
FreeSolv	Descriptor MLP	1.290 ± 0.446	0.732 ± 0.203	0.865 ± 0.077
	Dummy	3.619 ± 0.388	2.770 ± 0.353	-0.022 ± 0.046
	ECFP4 + RF	1.795 ± 0.306	1.157 ± 0.153	0.725 ± 0.133
	RDKit 2D + RF	1.089 ± 0.175	0.673 ± 0.106	0.903 ± 0.035
Lipophilicity	Descriptor MLP	0.683 ± 0.023	0.520 ± 0.016	0.642 ± 0.040
	Dummy	1.148 ± 0.039	0.923 ± 0.038	-0.006 ± 0.005
	ECFP4 + RF	0.853 ± 0.028	0.662 ± 0.023	0.443 ± 0.051
	RDKit 2D + RF	0.748 ± 0.025	0.581 ± 0.021	0.570 ± 0.047
BBBP	Descriptor MLP	0.915 ± 0.028	0.973 ± 0.014	–
	Dummy	0.500 ± 0.000	0.809 ± 0.031	–
	ECFP4 + RF	0.905 ± 0.022	0.970 ± 0.011	–
	RDKit 2D + RF	0.912 ± 0.019	0.970 ± 0.011	–
BACE	Descriptor MLP	0.787 ± 0.034	0.718 ± 0.042	–
	Dummy	0.500 ± 0.000	0.378 ± 0.051	–
	ECFP4 + RF	0.822 ± 0.028	0.751 ± 0.066	–
	RDKit 2D + RF	0.822 ± 0.027	0.730 ± 0.054	–

Table S10: Per-split primary test scores of the main models across the ten saved splits (RMSE for ESOL, FreeSolv and Lipophilicity; AUROC for BBBP and BACE).

Dataset	Model	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
ESOL	RDKit 2D + RF	0.6639	0.6808	0.6696	0.7547	0.6397	0.6375	0.6097	0.6183	0.5104	0.5848
	Chemprop graph	0.8309	0.8777	0.6733	0.8585	0.7594	0.7510	0.7892	0.7457	0.7504	0.8693
	Chemprop + x_d	0.6191	0.6783	2.7605	0.6756	0.5626	0.6685	0.6393	0.5890	0.6319	0.6157
	Framework graph	0.6463	0.5230	0.8418	0.7132	0.5693	0.6226	0.5806	0.5645	0.5293	0.6434
	Framework hybrid	0.6009	0.7098	0.6978	0.6768	0.5435	0.5941	0.6228	0.5466	0.6179	0.6161
FREESOLV	RDKit 2D + RF	1.2903	1.1880	1.1795	0.8323	1.0672	0.7108	1.1888	1.2401	1.1541	1.0420
	Chemprop graph	1.3274	1.2645	1.4407	0.8439	1.0739	0.9476	1.1504	1.9436	0.9271	0.8584
	Chemprop + x_d	1.1473	1.2149	1.2957	0.6556	1.1701	0.4933	0.7630	1.1099	0.8629	0.7455
	Framework graph	1.2610	1.2657	1.2055	1.4173	1.1968	1.0893	1.3345	1.4874	1.3265	0.9226
	Framework hybrid	1.4184	1.3344	0.9389	0.7654	1.3613	0.7925	1.0866	1.4291	0.9256	0.9384
Lipophilicity	RDKit 2D + RF	0.7626	0.7607	0.7486	0.7427	0.7339	0.7731	0.7235	0.7181	0.8006	0.7183
	Chemprop graph	0.5689	0.6142	0.6567	0.6265	0.6080	0.8845	0.5899	0.6023	0.6403	0.6141
	Chemprop + x_d	0.6471	0.6885	0.6364	0.6450	0.6021	0.8151	0.5592	0.6071	0.5977	0.5949
	Framework graph	0.5717	0.6222	0.6339	0.5716	0.5949	0.6373	0.5881	0.6000	0.5805	0.6686
	Framework hybrid	0.5922	0.6149	0.6173	0.5875	0.6277	0.6343	0.5708	0.6108	0.5877	0.6173
BBBP	RDKit 2D + RF	0.9161	0.9187	0.9305	0.9224	0.8680	0.8895	0.9102	0.9239	0.9334	0.9089
	Chemprop graph	0.9268	0.9047	0.9002	0.9253	0.9393	0.9062	0.9265	0.9204	0.8480	0.9120
	Chemprop + x_d	0.9459	0.9181	0.9412	0.9039	0.8897	0.9150	0.9563	0.9256	0.9222	0.9125
	Framework graph	0.9162	0.8265	0.9270	0.9243	0.9386	0.8850	0.9496	0.9413	0.9258	0.9097
	Framework hybrid	0.9367	0.9118	0.9507	0.9110	0.8958	0.8833	0.9659	0.8671	0.9280	0.9115
BACE	RDKit 2D + RF	0.8316	0.7946	0.8540	0.8259	0.8349	0.8604	0.8001	0.7731	0.8438	0.8065
	Chemprop graph	0.8288	0.6838	0.7933	0.8432	0.8343	0.8984	0.7864	0.7737	0.8615	0.7767
	Chemprop + x_d	0.8324	0.7522	0.8292	0.8077	0.8106	0.8614	0.7508	0.7444	0.7865	0.7500
	Framework graph	0.8147	0.7355	0.8017	0.8188	0.8125	0.8663	0.7699	0.7498	0.8309	0.8234
	Framework hybrid	0.8047	0.7892	0.8347	0.7904	0.7890	0.8252	0.7293	0.7465	0.7970	0.7481

Advisor case study (Caco-2)

The dataset advisor (`inspect`) was applied to the TDC Caco-2 permeability set (910 molecules; health score 95/100) and its recommendations were trained on one scaffold split. Every advisor configuration had a higher test RMSE than the plain D-MPNN baseline (Table S11). The advisor's rules are heuristics derived from dataset statistics, not from validation performance; its output is a starting configuration to be tested, which is why the main study evaluates design choices on matched splits rather than through the advisor.

Table S11: Caco-2 advisor case study (single scaffold split, 91 test molecules; 35 epochs, batch 32, patience 12).

Configuration	RMSE	MAE	R^2	Spearman	Best epoch
Baseline D-MPNN (MSE, no scheduler)	0.559	0.418	0.571	0.848	4
Advisor v1: + descriptors, Huber, cosine warm-up	0.593	0.443	0.518	0.843	5
Advisor v2: + virtual node, attention readout	0.628	0.473	0.460	0.828	4
Advisor v2 + pairwise ranking loss	0.634	0.468	0.450	0.837	4

Duplicate-controlled BBBP sensitivity analysis

Table S12: BBBP sensitivity analysis after excluding the 11 InChIKey groups with conflicting labels and retaining one record for each of the remaining 53 duplicated groups. The cleaned dataset contains 1,960 unique structures (1,497 positive and 463 negative). The same ten new scaffold manifests were used for all models. Scores are test AUROC. Because the dataset composition and split manifests differ from the primary benchmark, score changes should not be attributed solely to duplicate leakage.

Model	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	Mean ± SD
Framework graph only D-MPNN	0.8890	0.8527	0.8894	0.8554	0.9533	0.9528	0.8438	0.8097	0.7520	0.8578	0.8656 ± 0.0577
Framework hybrid with 210 descriptors	0.8953	0.8278	0.8668	0.8697	0.9074	0.8975	0.8953	0.8830	0.7642	0.8298	0.8637 ± 0.0422
Chemprop v2.3.1 graph only	0.8168	0.8251	0.8769	0.8655	0.9324	0.9155	0.9105	0.8833	0.8298	0.8305	0.8686 ± 0.0398
Chemprop v2.3.1 with 210 descriptors	0.8214	0.8384	0.8529	0.8706	0.9390	0.9358	0.9229	0.8927	0.8319	0.8378	0.8744 ± 0.0428
RDKit 2D + Random Forest	0.8804	0.8530	0.9195	0.8645	0.9332	0.9264	0.9092	0.9076	0.8414	0.9046	$\mathbf{0.8940 \pm 0.0305}$
Framework hybrid minus graph only (Δ AUROC)	+0.0063	-0.0249	-0.0227	+0.0142	-0.0459	-0.0552	+0.0515	+0.0733	+0.0122	-0.0279	-0.0019 ± 0.0393
Chemprop hybrid minus graph only (Δ AUROC)	+0.0045	+0.0134	-0.0240	+0.0051	+0.0066	+0.0203	+0.0124	+0.0094	+0.0022	+0.0073	$+0.0057 \pm 0.0111$

For the framework, the hybrid minus graph paired difference favoured the hybrid on 5/10 splits, with a descriptive paired 95% t interval of $[-0.0464, +0.0426]$. For Chemprop, it favoured the hybrid on 9/10 splits, with a corresponding interval of $[-0.0068, +0.0183]$. Both intervals include zero. The model ranking was not identical to the primary benchmark, and the score changes cannot be assigned to duplicate leakage alone because the dataset composition and split manifests changed.

Table S12b: Paired statistics for the duplicate-controlled BBBP sensitivity analysis (10 new scaffold manifests; the same manifests are used for every model). $\bar{\Delta}$ is A minus B in AUROC; p_{H} is Holm-adjusted within this table ($m = 5$). No comparison in this table survives Holm correction, so every entry is reported as inconclusive. Because the cleaned dataset and its split manifests differ from the primary benchmark, the ranking seen here cannot be attributed to duplicate leakage alone.

Comparison (A vs B)	$\bar{\Delta} \pm$ SD	Corrected 95% interval	p	p_{H}	Wins of A
Framework hybrid − Framework graph only	-0.0019 ± 0.0393	$[-0.0464, 0.0426]$	0.925	0.925	5/10
Chemprop hybrid − Chemprop graph only	$+0.0057 \pm 0.0111$	$[-0.0068, 0.0183]$	0.329	0.861	9/10
Framework hybrid − RDKit 2D + RF	-0.0303 ± 0.0289	$[-0.0629, 0.0024]$	0.065	0.327	2/10
Framework graph only − RDKit 2D + RF	-0.0284 ± 0.0426	$[-0.0766, 0.0198]$	0.215	0.861	3/10
Chemprop hybrid − RDKit 2D + RF	-0.0196 ± 0.0307	$[-0.0543, 0.0150]$	0.232	0.861	4/10

Checkpoint protocol and result-file inventory

The completed harmonised routing experiment supersedes the earlier ESOL-only checkpoint-rule sensitivity analysis. Every reported compact-route checkpoint is selected by minimum validation loss, and K^* is selected by the primary validation metric evaluated at that checkpoint. The file `benchmarks/results/routing_checkpoint_audit.csv` records the checkpoint-level audit across 250 model runs. The five dataset-specific JSON outputs preserve the split-level metrics and route selections. The files should be consulted for per-split paired analyses; Table S7 reports the cross-split mean and population standard deviation.

Table S13: Harmonised routing audit inventory. The checkpoint rule is minimum validation loss for all routes; the K^* selection metric is validation RMSE or validation AUROC, as appropriate. The filenames identify the dataset-specific split-level result records.

Dataset	Primary metric	K^* selection	Result file
ESOL	RMSE	Lowest validation RMSE at the minimum-validation-loss checkpoint	`evidence_routing_esol.json`
FreeSolv	RMSE	Lowest validation RMSE at the minimum-validation-loss checkpoint	`evidence_routing_freesolv.json`
Lipophilicity	RMSE	Lowest validation RMSE at the minimum-validation-loss checkpoint	`evidence_routing_lipophilicity.json`
BBBP	AUROC	Highest validation AUROC at the minimum-validation-loss checkpoint	`evidence_routing_bbbp.json`
BACE	AUROC	Highest validation AUROC at the minimum-validation-loss checkpoint	`evidence_routing_bace.json`